Linear Regression for Finance

From Ordinary Least Squares to Factor Models

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 4

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1 The Linear Model

Linear regression is the most fundamental tool in statistical modeling and quantitative finance. Nearly every factor model, risk model, and return-prediction framework is built on it. We begin with the general setup.

1.1 Model Specification

We observe n data points, each consisting of a response y_i and a vector of p predictors $(x_{i1}, x_{i2}, \ldots, x_{ip})$. The linear model posits that the response is a linear function of the predictors plus noise:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad i = 1, \ldots, n.$$

In matrix notation, we write this compactly as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where

- $\mathbf{y} \in \mathbb{R}^n$ — response vector;
- $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ — design matrix (with a column of ones for the intercept);
- $\boldsymbol{\beta} \in \mathbb{R}^{p+1}$ — parameter vector $(\beta_0, \beta_1, \ldots, \beta_p)^\top$;
- $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ — error vector.

Written out, the design matrix and vectors look like

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}_{\mathbf{y}} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}}_{\mathbf{X}} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}}_{\boldsymbol{\beta}} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{\boldsymbol{\varepsilon}}.$$

The first column of $\mathbf{X}$ is a column of ones, corresponding to the intercept β_0.

1.2 Assumptions

The results we derive in this note — unbiasedness, efficiency, the distribution of test statistics — depend on a set of assumptions about the model.

1. **Linearity.** The conditional mean of y given $\mathbf{X}$ is linear in the parameters: $E[y_i \mid \mathbf{X}] = \mathbf{x}_i^\top \boldsymbol{\beta}$.
2. **Strict exogeneity.** The errors have zero conditional mean: $E[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$. The predictors carry no information about the errors. In finance, this is often violated when we include lagged returns as predictors (endogeneity).
3. **Homoscedasticity and no autocorrelation.** Conditional on $\mathbf{X}$, the errors have constant variance and are uncorrelated: $\mathrm{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$. This is the *spherical errors* assumption: (i) each ε_i has the same variance σ^2, and (ii) $\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.
4. **Full rank.** The design matrix has full column rank: $\mathrm{rank}(\mathbf{X}) = p+1$. This rules out perfect multicollinearity. When it fails, $\mathbf{X}^\top \mathbf{X}$ is singular and $\hat{\boldsymbol{\beta}}$ is not uniquely defined.

Gauss–Markov Theorem. Under Assumptions 1–4, the OLS estimator $\hat{\boldsymbol{\beta}}$ (derived in Section 2) is the **Best Linear Unbiased Estimator (BLUE)**: among all estimators that are linear in $\mathbf{y}$ ($\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{y}$) and unbiased ($E[\tilde{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$), $\hat{\boldsymbol{\beta}}_{\mathrm{OLS}}$ has the smallest variance in the matrix sense:

$$\mathrm{Var}(\tilde{\boldsymbol{\beta}}) - \mathrm{Var}(\hat{\boldsymbol{\beta}}) \succeq \mathbf{0} \quad \text{(positive semidefinite)}.$$

Gauss–Markov tells us OLS is optimal within the class of linear unbiased estimators. If we are willing to accept some bias, we can potentially reduce variance and achieve lower mean squared error. This is precisely the idea behind Ridge regression and Lasso, covered in subsequent weeks.

In finance, Assumption 3 is the one that most frequently fails: volatility clustering violates the constant-variance assumption (heteroscedasticity). Return distributions also exhibit heavy tails, and the Jarque–Bera test almost always rejects normality for financial returns, which matters for the inference of Section 5. Diagnostic tests and remedies are discussed in Section 4.

2 Ordinary Least Squares

2.1 The OLS Objective

Given the model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, we want the parameter vector $\boldsymbol{\beta}$ that makes the model fit the data as closely as possible. The OLS criterion measures fit by the sum of squared residuals:

$$\min_{\boldsymbol{\beta}} S(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Expanding the quadratic form:

$$S(\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

2.2 Derivation of the OLS Estimator

To minimize $S(\boldsymbol{\beta})$, take the gradient with respect to $\boldsymbol{\beta}$ and set it to zero:

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{0} \quad \Longrightarrow \quad \mathbf{X}^\top \mathbf{X}\, \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}.$$

This is the **normal equation**. When $\mathbf{X}$ has full column rank (Assumption 4), $\mathbf{X}^\top\mathbf{X}$ is invertible, and

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} \tag{1}$$

This is the **OLS estimator**, the central formula of this note. The Hessian $\partial^2 S/\partial\boldsymbol{\beta}\,\partial\boldsymbol{\beta}^\top = 2\mathbf{X}^\top\mathbf{X}$ is positive definite under full rank, so $\hat{\boldsymbol{\beta}}$ is the global minimizer.

2.3 Geometric Interpretation

The fitted values

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$$

are the *orthogonal projection* of $\mathbf{y}$ onto the column space of $\mathbf{X}$, denoted $\mathcal{C}(\mathbf{X})$.

The hat matrix. Define $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$. Then

1. $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ (the hat matrix "puts the hat on $\mathbf{y}$");
2. $\mathbf{H}$ is symmetric: $\mathbf{H}^\top = \mathbf{H}$;
3. $\mathbf{H}$ is idempotent: $\mathbf{H}^2 = \mathbf{H}$;
4. $\mathrm{tr}(\mathbf{H}) = p + 1$ (the number of parameters).

These are the defining properties of a projection matrix.

2.4 Residuals

The residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}.$$

The matrix $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is also a projection matrix (symmetric, idempotent), projecting onto the orthogonal complement of $\mathcal{C}(\mathbf{X})$.

Residual orthogonality. The residuals are orthogonal to the column space of $\mathbf{X}$:

$$\mathbf{X}^\top\mathbf{e} = \mathbf{X}^\top\mathbf{y} - \mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0},$$

by the normal equation. Since the first column of $\mathbf{X}$ is $\mathbf{1}_n$, $\sum_{i=1}^n e_i = 0$: the residuals sum to zero whenever the model includes an intercept.

3 Statistical Properties of OLS

Throughout this section, we treat $\mathbf{X}$ as fixed (or condition on it) and let $\boldsymbol{\varepsilon}$ be the source of randomness.

3.1 Unbiasedness

Under Assumptions 1–2, the OLS estimator is unbiased: $E[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}$. Substituting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into (1):

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon},$$

$$E[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \underbrace{E[\boldsymbol{\varepsilon} \mid \mathbf{X}]}_{=\mathbf{0}} = \boldsymbol{\beta}.$$

3.2 Variance of the OLS Estimator

Under Assumptions 1–3, using $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}$:

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \underbrace{E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top \mid \mathbf{X}]}_{=\sigma^2\mathbf{I}_n}\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

The standard error of the j-th coefficient is therefore

$$\mathrm{se}(\hat{\beta}_j) = \sigma\sqrt{[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}}.$$

4 Diagnostics

After fitting a regression, check whether the assumptions of Section 1.2 hold. Violations can invalidate inference, inflate standard errors, or make predictions unreliable. The tools below are organized by the assumption they test.

4.1 Residual Plots

The simplest and most informative diagnostic is to plot the residuals:

1. **Residuals vs. fitted values** (e_i vs. $\hat{y}_i$). Should show no pattern. A funnel shape indicates heteroscedasticity; a curved pattern indicates non-linearity.
2. **Q-Q plot.** Residual quantiles against standard normal quantiles. Departures from the 45-degree line indicate heavy tails or skewness.
3. **Residuals vs. time** (time-series data). Should show no pattern. Serial correlation in residuals is very common in financial data.

4.2 Linearity (Assumption 1)

Ramsey RESET test. Augment the regression with powers of the fitted values ($\hat{y}^2, \hat{y}^3$) and test their joint significance via an F-test. Rejection suggests the linear functional form is misspecified.

4.3 Exogeneity (Assumption 2)

Hausman test. Compares OLS estimates with those from instrumental variables (IV/2SLS). A significant difference indicates endogeneity. Requires valid instruments, which are often hard to find in practice.

4.4 Heteroscedasticity (Assumption 3)

When $\mathrm{Var}(\varepsilon_i \mid \mathbf{X})$ is not constant, the OLS estimator is still unbiased but no longer efficient, and the standard errors are wrong.

- **Breusch–Pagan test.** Regress the squared residuals e_i^2 on the original predictors $\mathbf{X}$. Under the null of homoscedasticity, $nR^2_{\text{aux}} \sim \chi^2_p$. A small p-value indicates heteroscedasticity.
- **White test.** A more general version that also includes squares and cross-products of the predictors in the auxiliary regression. It requires fewer assumptions and can detect more forms of heteroscedasticity.

Remedy: use **heteroscedasticity-consistent (HC) standard errors**, also called White or robust standard errors.

4.5 Autocorrelation (Assumption 3)

When the errors ε_t are serially correlated — as is common in financial time series — the OLS standard errors are unreliable. The **Durbin–Watson statistic** is

$$DW = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2} \approx 2(1 - \hat{\rho}),$$

where $\hat{\rho}$ is the first-order autocorrelation of the residuals. $DW \approx 2$ indicates no autocorrelation; $DW \ll 2$ positive autocorrelation; $DW \gg 2$ negative autocorrelation. The statistic is only meaningful when the data have a natural ordering.

Remedy: use **Newey–West (HAC) standard errors**, which are robust to both heteroscedasticity and autocorrelation.

4.6 Multicollinearity (Assumption 4)

When predictors are highly correlated, $\mathbf{X}^\top\mathbf{X}$ is nearly singular. The OLS estimates are still unbiased, but their variances become very large: small changes in the data lead to large changes in $\hat{\boldsymbol{\beta}}$.

- **Variance inflation factor.** For the j-th predictor, $\mathrm{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 from regressing x_j on all other predictors. Rule of thumb: $\mathrm{VIF}_j > 10$ indicates serious multicollinearity.
- **Condition number** of $\mathbf{X}$: values above ∼30 indicate problematic collinearity.

In factor models, multicollinearity among factors is a real concern: the Fama–French "size" and "value" factors have non-trivial correlation. Regularization (Ridge, Lasso) is one remedy; orthogonalizing the factors is another.

4.7 Normality of the Errors

Normality is not required for the OLS estimator or for Gauss–Markov, but exact finite-sample inference (Section 5) relies on it. The **Jarque–Bera test** uses

$$JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2_2 \quad \text{under } H_0,$$

where S is the skewness and K the kurtosis of the residuals. The Q-Q plot gives the same information visually.

5 Inference

Assumptions 1–4 give the mean and variance of $\hat{\boldsymbol{\beta}}$ but not its distribution, which we need for hypothesis tests and confidence intervals. By the Central Limit Theorem, $\hat{\boldsymbol{\beta}}$ is approximately normal for large n, so the tests and intervals below are approximately valid. They are exact in finite samples if, in addition, $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

5.1 Distribution of $\hat{\beta}$

Under Assumptions 1–4, for large n:

$$\hat{\boldsymbol{\beta}} \overset{a}{\sim} \mathcal{N}\big(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\big), \qquad \hat{\beta}_j \overset{a}{\sim} \mathcal{N}\big(\beta_j, \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}\big).$$

In practice σ^2 is replaced by $s^2 = \mathbf{e}^\top \mathbf{e}/(n-p-1)$.

5.2 t-Tests for Individual Coefficients

The most common test is whether a single coefficient differs from zero, $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$. The test statistic is

$$t_j = \frac{\hat{\beta}_j}{\widehat{\mathrm{se}}(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{s\sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}}$$

Under H_0, $t_j \sim t_{n-p-1}$. Reject at level α if $|t_j| > t_{n-p-1,\alpha/2}$. Rule of thumb: $|t| > 2$ roughly corresponds to significance at the 5% level when n is moderately large.

5.3 Confidence Intervals

A $(1-\alpha) \times 100\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{n-p-1,\alpha/2} \cdot \widehat{\mathrm{se}}(\hat{\beta}_j),$$

approximately $\hat{\beta}_j \pm 1.96 \cdot \widehat{\mathrm{se}}(\hat{\beta}_j)$ for a 95% interval with large n.

5.4 F-Test for Overall Model Significance

To test whether *all* slope coefficients are simultaneously zero, $H_0 : \beta_1 = \cdots = \beta_p = 0$ vs. H_1: at least one $\beta_j \neq 0$:

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)} = \frac{R^2/p}{(1-R^2)/(n-p-1)}$$

where $\mathrm{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$ is the total sum of squares. Under H_0, $F \sim F_{p,\, n-p-1}$.

5.5 R^2 and Adjusted R^2

The coefficient of determination measures the fraction of variance explained:

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \in [0, 1].$$

R^2 never decreases when predictors are added, even irrelevant ones. The **adjusted** R^2 penalizes model complexity,

$$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-p-1},$$

and can decrease when a predictor does not sufficiently improve the fit.

In finance, R^2 values for return-prediction regressions are typically very low — even 1–2% out-of-sample can be economically meaningful. What matters is whether the predictions translate into profitable trading strategies after transaction costs.

6 Summary

Quantity	Formula
OLS estimator	$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
Fitted values	$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$
Residuals	$\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$
Variance of $\hat{\boldsymbol{\beta}}$	$\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$
σ^2 estimator	$s^2 = \mathbf{e}^\top \mathbf{e} / (n - p - 1)$
t-statistic	$t_j = \hat{\beta}_j \,/\, [s \sqrt{(\mathbf{X}^\top \mathbf{X})^{-1}_{jj}}]$
R^2	$1 - \mathrm{RSS}/\mathrm{TSS}$
Adjusted R^2	$1 - (1 - R^2)(n-1)/(n-p-1)$

The takeaway. OLS is a projection: under Assumptions 1–4 it is unbiased and BLUE, and its standard errors follow from $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. In financial data Assumption 3 routinely fails, so the point estimates survive but the default standard errors do not — use HC or HAC errors before reading a t-statistic.